MACQUARIE University
SYDNEY·AUSTRALIA

COMP8221 Advanced Computer Vision

Assignment 1: Diffusion Models

Design and Evaluation of a DDIM-Based Diffusion Model for Image Generation

Student Name: Md Ifthekher Uddin Chy

Student ID: 48546356

Unit Convenor: Prof. Longbing Cao

May 2, 2026

Contents

1 Introduction **2**

2 Methodology **2**

2.1 Overview of Diffusion Models 2

2.2 DDIM 3

2.3 Model Architecture 4

2.4 Objective Function 5

3 Training **5**

3.1 Dataset and Preprocessing 5

3.2 Training Procedure 6

4 Evaluation **6**

4.1 Loss Curve 6

4.2 FID Evaluation 7

5 Analysis and Visualization **8**

5.1 Real Dataset Samples 8

5.2 Forward Diffusion Process 9

5.3 Generated Samples 9

5.4 Reverse Diffusion Process 10

5.5 Qualitative Discussion 11

6 Limitations **11**

7 Conclusion **12**

Section 1 Introduction

Generative models are an important field in machine learning, particularly for generating images. They aim to capture the data distribution and then create new samples that resemble the original data. Generative models can be of various types including autoregressive models, variational autoencoders (VAEs) and diffusion models. Each has its own way of doing it.

In this project, this work focuses on diffusion models. Diffusion models have been very successful in producing high-quality images compared to other methods. They do not directly generate the image, but rather transform noise into a sample. This makes them more robust to train and they are able to generate better images. The overall concept is inspired by the research on denoising diffusion probabilistic models [1].

I chose Option 3 of creating a generative model based on diffusion for this assignment. In particular, I employed Denoising Diffusion Implicit Models (DDIM). DDIM is a modification of the original diffusion model that can sample images faster without compromising too much on image quality [2]. This was necessary because the project is run on limited computational resources, so faster sampling makes the model more practical.

In this project, the aim is to create an entire diffusion model. This involves the forward noising process, the reverse sampling process and the neural network that estimates the noise at each step. The model is trained on the CIFAR-10 dataset [3], which is a popular dataset for image generation.

In summary, the goal of this project is to build the complete DDIM pipeline and test how well it can generate images from arbitrary noise. The key focus here is not on reaching the best performance, but rather to build the model properly, understand how it works and analyse its behaviour through both quantitative and qualitative results.

Section 2 Methodology

2.1 Overview of Diffusion Models

Diffusion models are a type of generative model that works by learning how to progressively add noise to an image, and then learns to undo the process. Rather than generating the image directly, it starts with a clean image and progressively adds noise until it is nearly completely obscured. This is known as the forward diffusion process. A tiny bit of Gaussian noise is added at each step, making it a gentle transformation.

Once the image is almost completely noise, the model learns to do the opposite, which is to start with noise and reconstruct the image. The innovation here is that the model does not try to directly generate the image, but rather the noise added at each step.

Then, the estimated noise can be removed step-by-step to get an image. This method is used in denoising diffusion probabilistic models (DDPMs) [1].

The forward process is predetermined and does not need to be learned. It has a fixed schedule that determines the amount of noise at each step. However, the backward process is learned with a neural network. The model is trained by providing it with a noisy image and a timestep, and it outputs the noise that is present. Through training, the model learns to recognise different types of noise and how to eliminate it.

A key benefit of diffusion models is that the training process is more stable and robust than other types of generative models. Because the model is predicting the noise, rather than directly generating the image, it does not suffer from some of the problems that GANs face, such as mode collapse. However, a drawback is that the generation process can be slow because it involves multiple steps to generate an image from noise.

In this project, this entire process is implemented using a predetermined schedule for the forward diffusion and a learned reverse process using a neural network. This can be observed in the forward noising visualisation (forward_noising_grid.png), where more noise is added to the image as the timestep progresses.

2.2 DDIM

Denoising Diffusion Implicit Models (DDIM) is a modification of the original diffusion process that alters the reverse process. The standard diffusion approach (DDPM) uses a stochastic process to denoise an image, but DDIM does this in a more deterministic manner. It allows us to generate images with fewer steps at a reasonable level of quality [2].

The standard diffusion process involves some randomness in each step of the reverse process, and so the model has to sample a distribution at each timestep. This is modified in DDIM by eliminating the randomness by setting the parameter η to 0. This makes the reverse process deterministic and the generated image will be the same when given the same initial noise. This allows DDIM to be more controllable and efficient when sampling.

The efficiency of DDIM is a key advantage in this project. The basic diffusion models can take hundreds or even thousands of steps to generate an image, which can be inefficient, particularly on weaker computing resources. DDIM can significantly reduce this number of steps. Here, we use only 50 sampling steps (instead of the original 1000 training timesteps) to generate new images, which allows a trade-off between speed and image quality.

It is also worth noting that DDIM uses the same model for predicting noise as DDPM. The only change is in the reverse sampling process. This allows the model to be trained

without any changes and the improved sampling can be achieved without modifying the model structure.

In this project, DDIM is used as the primary sampling method after training. The network is trained to predict the noise at each timestep, and DDIM samples images from noise using this prediction. This enables the model to produce images faster while maintaining the diffusion process.

2.3 Model Architecture

The neural network that is the focus of the diffusion model in this project is a U-Net that is used to estimate noise at every timestep. The model does not produce images directly but rather learns to approximate the noise that has been introduced to an image and to reconstruct the image gradually. It is a common architecture employed in the context of diffusion models since it is able to learn both local and global features in an effective manner [4].

The U-Net architecture is based on an encoder-decoder structure. The input image is downsampled in a sequence of convolutional layers in the encoder part. This assists the model to learn higher-level features and comprehend the general structure of the image. The feature maps are upsampled back to the original image size in the decoder part. This enables the model to reconstruct spatial information and generate outputs which have the same resolution as the input.

Skip connections are one of the significant elements of U-Net. These links interconnect the encoder and decoder at various levels, enabling the model to reuse feature information from earlier layers. This is used to maintain small details that could be lost during downsampling. The absence of these connections would likely result in poor output images.

The other essential component of the model is the time embedding. As the model needs to know the timestep at which the image is being processed, a timestep value is provided as input. This timestep is then represented in higher dimensions using sinusoidal embeddings, similar to those used in transformer models. This embedding is then incorporated into the network so that it can adapt its predictions based on the amount of noise.

In this application, the U-Net is relatively lightweight compared to large-scale diffusion models, mainly because of computational constraints. With this simplified configuration, it can learn noise patterns of the data and support the reverse diffusion process. The model is given a noisy image and a timestep as input and generates a prediction of the noise, which is then used in the DDIM sampling process.

2.4 Objective Function

The aim of the diffusion model in this project is to learn to predict the noise that is added to an image at a particular timestep. The model does not explicitly produce images, but rather concentrates on estimating the noise component, which can then be removed step by step to recover the original image. This method makes learning easier and has been shown to be useful in diffusion-based models [1].

During training, a clean image is sampled from the dataset and noise is added to it following a predetermined schedule. A timestep is also randomly selected, which determines the amount of noise added. This noisy image together with the timestep is then given as input to the model, and the model attempts to predict the exact noise that was added.

A mean squared error (MSE) loss between the predicted noise and the actual noise is used to measure the performance of the model. This loss encourages the model to estimate the noise correctly at different noise levels. Over time, the model learns the behaviour of noise across various timesteps, which allows it to generalise well during the reverse process.

A benefit of this objective is that it stabilises the training compared to directly predicting the clean image. Since the noise follows a known distribution, the model is able to learn a simpler and more structured task. This also helps the model remain consistent across different stages of the diffusion process.

In this project, MSE is used as the loss function to predict the noise, and it is applied at random timesteps during training. This enables the model to learn to handle both low-noise and high-noise conditions, which is important for generating images effectively during the reverse diffusion process.

Section 3 Training

3.1 Dataset and Preprocessing

The dataset utilized in this project to train and evaluate is CIFAR-10 dataset [3]. It is composed of 32x32 RGB pictures in 10 classes. This dataset is very popular to do image generation works and it can be applied in this assignment due to its small size and easy complexity.

The images are preprocessed before training to fit into the needs of the diffusion model. The images are all normalised to the range [-1, 1], which works well in training and adding noise consistently. Moreover, horizontal flipping is done randomly in the training process to amplify the data variation to a relatively small degree.

The data are divided into training, validation, and test sets. The noise prediction model is learned with the training set and performance is monitored with the validation set. The test set is primarily utilized in the evaluation process, especially in the calculation of metrics such as FID.

PyTorch DataLoaders are employed to process the data with the help of batching. This enables the model to work with multiple images simultaneously, and enhances the training effectiveness. In general, the preprocess steps are maintained minimal yet adequate to train a diffusion model on CIFAR-10.

3.2 Training Procedure

In this project, the model is trained with a typical diffusion model training setup. The optimizer employed is AdamW with a learning rate of 0.0002 that gives consistent convergence in training. The model is trained over a specified number of epochs with a batch size of 64 as a tradeoff between computational efficiency and memory consumption.

In training, a timestep is sampled at random for each input image. Noise is introduced to the image, based on this timestep, with the predefined diffusion schedule. The model then receives the noisy image and the timestep as input and makes a prediction of the noise added. This is repeated across different timesteps so that the model learns to deal with the varying noise levels.

A validation step is also part of the training process, in which every epoch is followed by the evaluation of the model on a validation set. This aids in monitoring the proper learning of the model and also aids in checking any overfitting. The validation losses and training losses are recorded and used later to analyze.

In order to have reproducibility, a fixed random seed is employed in the course of the experiment. The training is done using hardware available with a GPU acceleration where possible to minimize training time. In general, the configuration is maintained simple and efficient to train a diffusion model on CIFAR-10.

Section 4 Evaluation

4.1 Loss Curve

The loss curve is used to monitor the training process and indicates how the model performance varies with time. The loss during training and validation is also computed at every epoch in this project. These values offer an easy method through which one can verify that the model is learning as intended.

Based on the loss curve one can see that the training loss progressively reduces with

the increase in the number of epochs. This means that the model is learning to predict the noise more precisely with time passing. The error on validation is of similar nature and this indicates that the model is generalising quite well to unseen data.

No high indications of instability during training, e.g. abrupt changes in the values of the loss. This indicates that the training process is stable. As the number of epochs during which the model is trained is limited, the loss does not decrease to a very low value, yet it still demonstrates a steady improvement.

Figure 1: Training and validation loss curve

In general, the loss curve shows that the model can learn the noise prediction task successfully. It also offers a handy understanding of the training behaviour without the need to evaluate the training behaviour with sophisticated metrics.

4.2 FID Evaluation

The Frechet Inception Distance (FID) is employed to measure the quality of the produced images. FID is a popular metric of generative models, which quantifies the discrepancy between the distribution of real images and generated images [5]. The smaller the FID score, the closer the generated images are to the real data distribution.

In this project, the FID is calculated by comparing features of real CIFAR-10 images and model-generated images. The features are extracted using a pretrained Inception network and the distance between the two distributions is evaluated using the statistics of the two distributions.

The FID score obtained in this experiment is 1.647053 using 64 generated samples. The model has a reasonable FID score considering the small training configuration and the computing limitations. Notably, the analysis is conducted on a comparatively limited

number of generated samples and this can impact the stability of the measure. Consequently, the reported FID score cannot be considered an accurate performance measure, but rather as a rough estimate.

Altogether, the FID assessment is a promising quantitative tool to supplement the qualitative findings. It can be used to assure that the model can produce images which are structurally similar to real images though the training is limited.

Section 5 Analysis and Visualization

5.1 Real Dataset Samples

The actual samples of data give us an idea of what is supposed to be learnt by the model. The CIFAR-10 pictures have a blend of various objects in terms of colour, texture, and shape. Although the resolution is small, there remains sufficient variation to be useful to the model. These samples assist in the comparison of the closeness of the generated outputs to the original data.

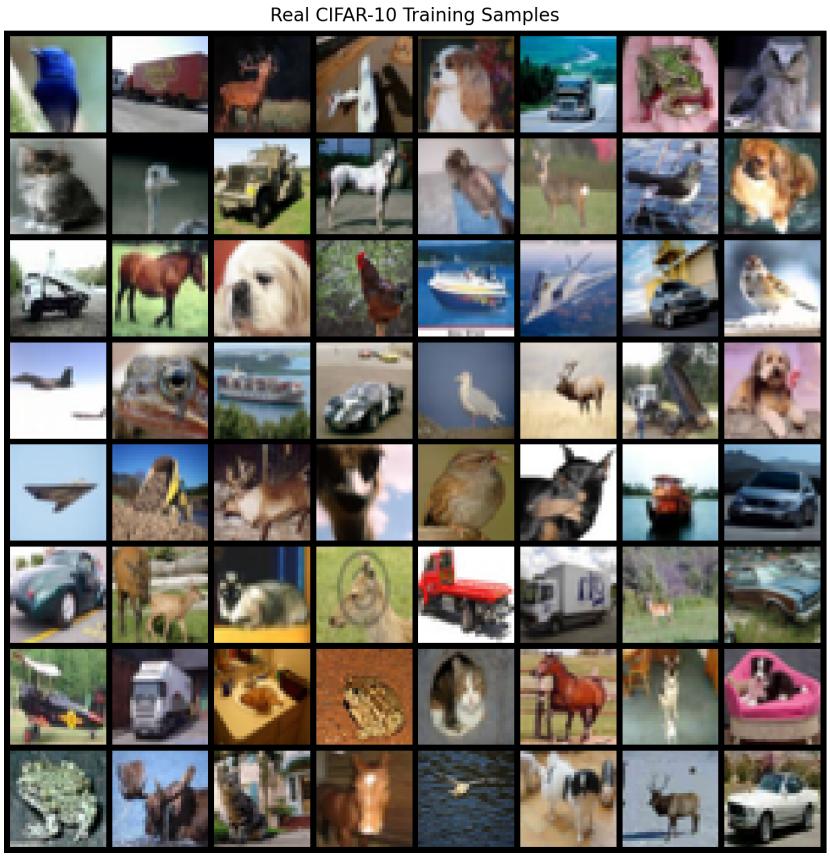

Figure 2: Real CIFAR-10 samples

5.2 Forward Diffusion Process

The forward diffusion process is visible in the Figure 3. With increase in the timestep, additive noise is added to the image until it is nearly all random. The conversion between a clear image and noise is gradual indicating that noise schedule is functioning properly. This is necessary as it determines what the model needs to learn in the course of training.

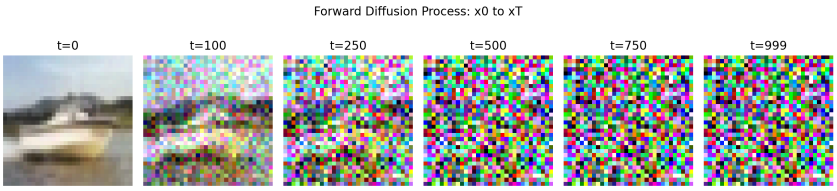

Figure 3: Forward diffusion process

5.3 Generated Samples

The obtained samples indicate the result of the model with pure noise as the input. The model can generate images, post-training, that are approximate in their similarity to the structure of the CIFAR-10 dataset. Other photos display simple shapes and colour patterns, although it might be a little blurry or noisy. This is not surprising because of poor training and model size, but still indicates that the model has learnt the reverse diffusion process to a point.

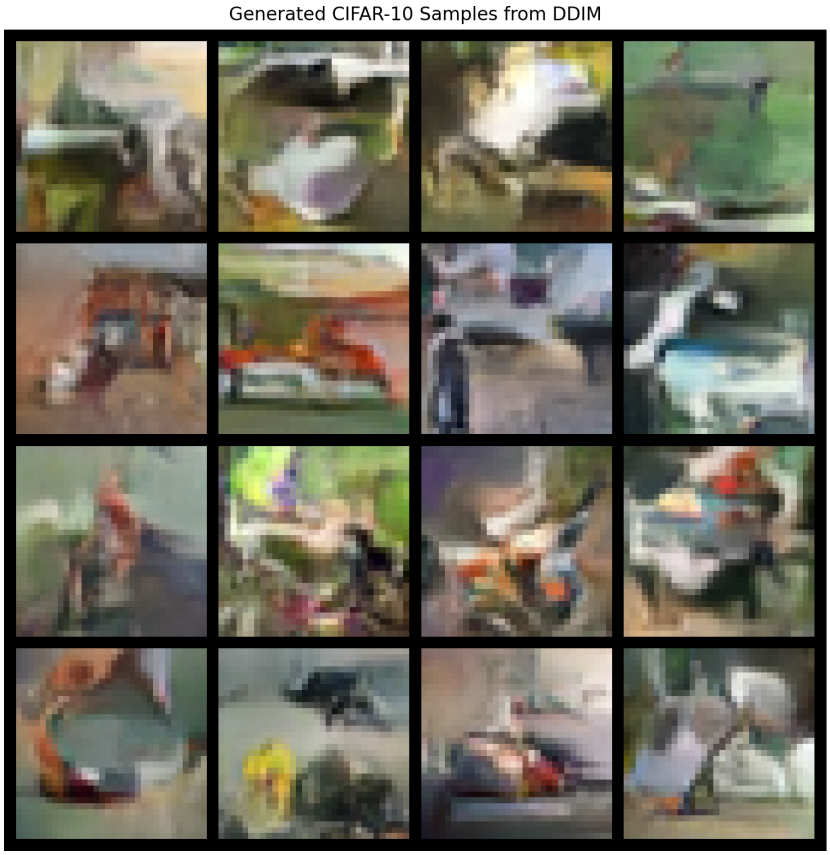

Figure 4: Generated samples from the model

5.4 Reverse Diffusion Process

In the reverse diffusion process, the behaviour can be observed in Figure 5. The image begins with pure noise and is refined in each step to eventually create a structured output. The fact that noise is replaced by a more meaningful image indicates that the model has the capability of eliminating noise in a gradual manner using the learned predictions. The steps in between may still look unclear, but the general trend of the process indicates that the reverse sampling is being done as intended.

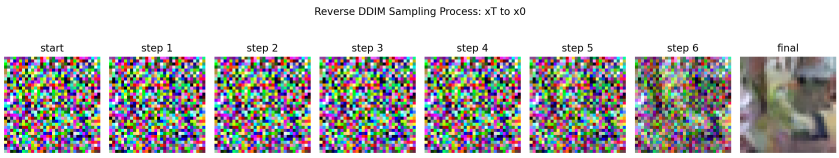

Figure 5: Reverse diffusion process

5.5 Qualitative Discussion

The images generated are somewhat faithful and diverse, qualitatively speaking, despite the constrained training. The model can generate variations of colour and simple shapes and this indicates that it has acquired some underlying patterns of the data set. But the artifacts that still exist include blur, noisy patches and blurred object boundaries.

The trade-off presented by DDIM is one of the key aspects. The generation can be made faster by decreasing the number of sampling steps but can have a minor impact on the image quality. In this implementation 50 sampling steps give a fair compromise between speed and output quality. The visual results can be further enhanced with additional training and potentially more sampling steps.

In general, quantitative findings are complemented by the qualitative results. Although the model is not completely trained, it can still exhibit the fundamental behaviour of a diffusion model, such as creating images based on noise and refining them stepwise.

Section 6 Limitations

This project has a number of limitations, which are primarily caused by computational limitations and the scope of the assignment. The model is not trained over many epochs which implies that it is probably undertrained. Consequently, the resulting images can still be noisy or not clearly structured in most instances.

Another constraint is the dataset. CIFAR-10 is full of low-resolution images (32×32) and this limits the amount of detail that the model can learn and create. Although it is appropriate to experiment, it is not representative of the high-resolution performance of diffusion models.

It also has a limited evaluation using FID. The score is calculated based on a comparatively low number of generated samples thus making the result less stable and less reliable. More samples would give a more precise estimate of the performance of the model.

Moreover, the model architecture adopted in this project is also quite simple in contrast to the large-scale diffusion models of the modern days. This has an impact on the quality of the produced outputs. More complicated architectures and extended training may considerably enhance the results.

In general, all these drawbacks do not influence the accuracy of the implementation, but they influence the quality of the outputs. The primary goal of this project is to show a functional DDIM-based diffusion model and not the state-of-the-art performance.

Section 7 Conclusion

A diffusion model based on DDIM is used in this project to create images by using noise. The model is constructed based on a U-Net architecture and trained to forecast noise at various timesteps. The forward diffusion process and the reverse sampling process are both successfully implemented.

The findings demonstrate that the model can learn the overall behaviour of the data and produce images that to a certain degree resemble the CIFAR-10 data. Although the quality is constrained by the limitations of the training, the outputs nonetheless illustrate the main concept of diffusion-based generation.

The performance is analysed by both qualitative and quantitative assessments. The samples and reverse diffusion process generated demonstrate the way the model converts noise into structured images. The FID score provides a numerical measure, but it is limited by the number of samples.

In general, the implementation fulfills the primary goal of the assignment. It illustrates an example of a working diffusion model with DDIM, and a full training and evaluation pipeline. Although it could have been improved, the project offers a clear picture of the way diffusion models can be utilized in practice.

References

[1] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[2] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," *International Conference on Learning Representations (ICLR)*, 2020.

[3] A. Krizhevsky, "Learning multiple layers of features from tiny images," University of Toronto, Tech. Rep., 2009.

[4] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2015.

[5] M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter, "Gans trained by a two time-scale update rule converge to a local nash equilibrium," *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.